

A Product Requirements Document (PRD) is the **foundational strategic blueprint for any product initiative**, especially innovative ones. It articulates the crucial **why**, **what**, and **for whom** of the product, aligning diverse teams towards a common goal. A well-crafted PRD specifies product functionalities, features, and success criteria, ensuring the final product meets customer needs and drives business growth. For innovative products, it is particularly important to capture the dynamic, iterative nature of development, which can be challenging with traditional text-heavy documents. **Mind maps are presented as indispensable visual tools for injecting clarity, fostering shared understanding, visualizing complex relationships, and managing risks associated with innovation directly within the PRD framework.**

Here is a blueprint template for a PRD, drawing on the essential mind maps and information from the sources:

Product Requirements Document (PRD) Blueprint

I. Introduction

- **1.1 Purpose:** SafeRoute AI transforms fragmented road safety data into predictive intelligence that helps Bengaluru authorities prevent night-time accidents by identifying high-risk segments and recommending targeted interventions.
- **1.2 Goals and Objectives:** Reduce night-time road fatalities by 25% through data-driven intervention prioritization, provide explainable risk assessments for 500m road segments, and enable evidence-based safety resource allocation.
- **1.3 Scope:** The hackathon prototype covers Bengaluru's road network with focus on night-time risk prediction, integrating BTP crash data with infrastructure mapping, and delivering a functional What-If Safety Simulator for intervention comparison.
- **1.4 Target Audience (for the PRD):** Primary users include traffic police commissioners, urban planning engineers, and road safety policy analysts who need actionable intelligence for accident prevention and infrastructure improvement decisions.

II. Product Vision and Strategy

- **2.1 Product Vision:** To become India's leading predictive road safety platform that transforms how cities prevent accidents by shifting from reactive response to proactive risk mitigation through AI-powered intelligence.
- **2.2 Strategic Objectives:** Establish SafeRoute AI as the standard for urban road safety decision-making across Indian cities, achieve 80% prediction accuracy for severe crashes, and demonstrate measurable accident reduction in pilot deployments.
- **2.3 Core Value Proposition:** SafeRoute AI uniquely combines predictive risk scoring with explainable AI and intervention simulation, enabling authorities to answer 'what to fix first and why' rather than just 'where accidents happened'.
- **2.4 Key Differentiators:** Unlike reactive hotspot mapping, SafeRoute AI provides forward-looking 0-100 Safety Scores with confidence levels, explains specific risk factors per segment, and quantifies expected safety impact of different interventions before implementation.

III. Target Audience and User Understanding

- **3.1.1 User Persona: Traffic Police Commissioner:**
Goal: Allocate limited enforcement resources to high-risk night-time corridors where intervention will prevent the most serious accidents and save lives.
Journey: Reviews dashboard showing top 10 highest risk segments with Safety Scores and predicted fatality likelihood, Examines risk factor breakdown identifying speeding, poor lighting, and junction complexity as primary causes, Uses WhatIf Simulator to compare speed enforcement versus lighting improvement and selects intervention with highest safety ROI
- **3.1.2 User Persona: Urban Planning Engineer:**
Goal: Design infrastructure improvements for dangerous road segments based on data-driven understanding of specific risk factors and vulnerable user patterns.

Journey: Identifies highrisk segments on planned development corridors and analyzes junction density, crossing safety, and bus stop proximity, Reviews historical accident patterns showing 65% motorcyclist involvement and peak risk between 9 PM12 AM, Simulates junction redesign and safer crossing interventions to validate design choices before construction

- **3.1.3 User Persona: Road Safety Policy Analyst:**

Goal: Develop evidence-based safety policies and measure intervention effectiveness across Bengaluru's road network to inform state-wide safety strategy.

Journey: Analyzes citywide risk distribution patterns identifying that 15% of segments account for 60% of predicted severe crashes, Compares intervention effectiveness across different road types showing lighting improvements reduce risk by 35% on arterial roads, Tracks beforeafter Safety Score changes on treated segments to validate model predictions and refine policy recommendations

- **3.2 User Needs and Expectations:** Users need actionable intelligence that goes beyond accident statistics to predict where serious risk exists, explain why specific segments are dangerous, and quantify which interventions will be most effective before committing resources.
- **3.3 Context of Use:** The platform is used in office settings for strategic planning and resource allocation decisions, with mobile access for field verification of high-risk segments and real-time intervention monitoring during night-time enforcement operations.

IV. User Journey and Experience

- **4.1 Key User Scenarios:** A traffic commissioner receives alert about newly identified high-risk corridor, reviews the Safety Score and risk factors, compares three intervention options in the simulator, and deploys the recommended action within 48 hours.
- **4.2 User Journey Maps:** Users start with city-wide risk overview, drill down to specific segments, explore explainable risk factors with visual breakdowns, simulate interventions with predicted impact scores, and export prioritized action plans with supporting evidence.
- **4.3 Pain Points and Opportunities:** Current pain points include fragmented data sources requiring manual integration, lack of predictive capability forcing reactive responses, and inability to compare intervention effectiveness before implementation, all addressed by SafeRoute AI's unified predictive platform.

V. Product Features and Functionality

- **5.1 Feature Breakdown:** Core features include 0-100 Safety Score calculation for 500m segments, explainable risk factor analysis showing contribution percentages, vulnerable user identification, What-If Safety Simulator with intervention comparison, and interactive map dashboard with filtering and export capabilities.
- **5.2 Functional Requirements:** System must process BTP crash data and OpenStreetMap infrastructure within 24 hours of updates, generate risk predictions with 80% accuracy, explain top 5 risk factors per segment, simulate 4 intervention types, and support 50 concurrent users with sub-2-second response times.
- **5.3 Non-Functional Requirements:** Platform must maintain 99.5% uptime during night hours (6 PM-2 AM), ensure data privacy compliance for accident records, provide mobile-responsive interface, support offline report generation, and scale to analyze 10,000+ road segments across Karnataka.

VI. Market Analysis and Competitive Landscape

- **6.1 Market Opportunity:** India's road safety market represents \$2B annual opportunity with 150,000+ deaths yearly, while government initiatives like Motor Vehicle Amendment Act 2019 mandate data-driven safety interventions, creating demand for predictive intelligence platforms.
- **6.2 Competitive Analysis:** Existing solutions like iRAP provide reactive risk mapping without prediction, while Zendrive focuses on driver behavior rather than infrastructure risk, leaving SafeRoute AI uniquely positioned with predictive, explainable, intervention-focused intelligence for government decision-makers.